

Lecture 7

Thurs AM 7/16

Outline / topics

- Off-momentum motion
 - ↳ closed orbit / dispersion
 - ↳ Relationship to momentum compaction
- Dispersion matching
- Chromatic aberration
 - ↳ chromaticity + correction
 - ↳ relationship to tune + resonances
- Real magnets
 - ↳ dipole edge example

Off-momentum motion

Lee 2.IV.1

- Finally time to consider $p \neq p_0$!
- First: think of impact on closed orbit, then on betatron oscillations

Recall we started with: (Lee 2)

$$x'' - \frac{\rho+x}{\rho^2} = \pm \frac{B_z}{B\rho} \left(1 + \frac{x}{\rho}\right)^2 \left(\frac{p_0}{p}\right)$$

+ charge
- charge

previously $p = p_0$
Recall $B\rho = \frac{p}{q}$

For linear $B_z \Rightarrow B_0 + B'_x x$ quad. gradient
 $\approx$ dipole

Write in the form for $p = p_0$

$$x'' + K_x(s)x = 0 \quad z'' + K_z(s)z = 0$$

$$K_x = \frac{1}{\rho^2} + K_1(s) \quad K_1(s) = \mp \frac{B_1(s)}{B\rho}$$

bending
(dipole)

focusing
strength
of quad. gradients

Add of t -momentum perturbation:

$$\delta = \frac{\Delta p}{p} \quad \text{where } \Delta p = p - p_0$$

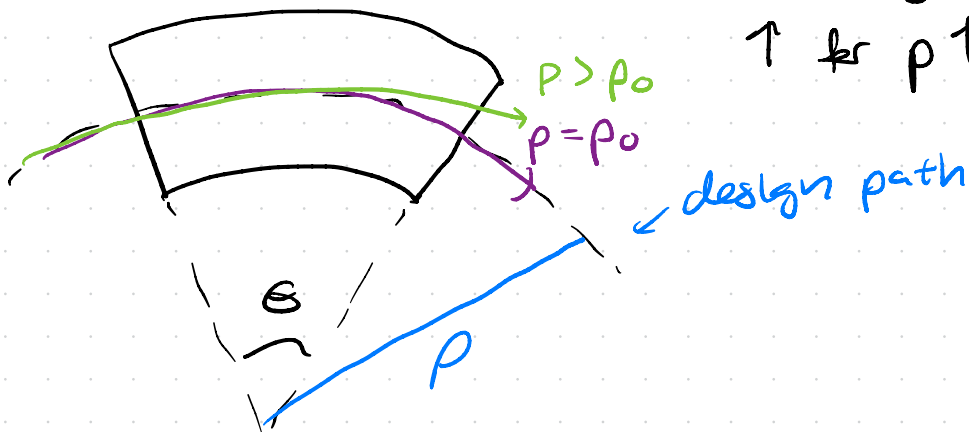
$\uparrow$
fractional
momentum
deviation

δ small, $\sim 10^{-4} - 10^{-2}$
 $\Rightarrow$ perturbation

- Before jumping into ECM, think conceptually about particle on the design closed orbit

* Bending radius

$\uparrow$ for $p \uparrow$



Effect is a kick in its trajectory

$\hookrightarrow$ closed orbit changes

$\hookrightarrow$ betatron oscillations change

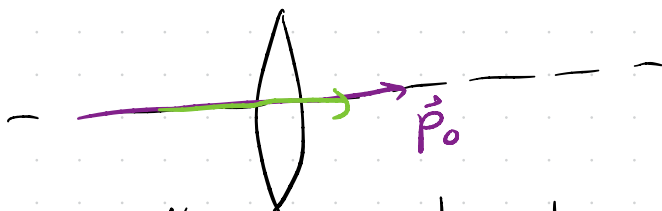
- Describe how it impacts coordinates:

$$x(s) = x_\beta(s) + D(s)\delta$$

$\uparrow$ betatron oscillation

$D(s)$ = dispersion function, how $x(s)$ changes with δ

- Dispersion introduced by dipoles, result of bending
- Quadrupoles + drifts don't introduce dispersion to closed orbit



But once dispersion has been introduced, affects motion through quads + drifts

$\hookrightarrow$ describe by propagating $D(s)$

- Impact on betatron oscillations around closed orbit treated separately $\Rightarrow$ chromaticity

- Describe general perturbed Hill eq:
H.W. S.5.4

$$U'' + K(s)U = K(s)\delta$$

neglecting $O(\delta^2)$ and higher

- Inhomogeneous eq: solution that is homogeneous sol + particular sol
 x_p above

$$U(s) = \underbrace{a C(s) + b S(s)}_{\substack{\uparrow \\ \text{cos-like} \quad \uparrow \\ \text{sin-like}}} + \delta D(s)$$

$$U'(s) = a C'(s) + b S'(s) + \delta D'(s)$$

$$\Rightarrow \begin{bmatrix} U(s) \\ U'(s) \\ \delta \end{bmatrix} = \begin{bmatrix} C(s) & S(s) & D(s) \\ C'(s) & S'(s) & D'(s) \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U_0 \\ U'_0 \\ \delta \end{bmatrix}$$

Back to

$$X'' + K_x(s)X = 0 = X'' + \left(\frac{1}{\rho^2} + K(s)\right)X = 0$$

↑
breaking up dipole
quad poles

expansion in δ ,

linear terms only, perturbed form is

$$X'' + \left(\frac{1-\delta}{\rho^2(1+\delta)} + \frac{K(s)}{(1+\delta)}\right)X = \frac{\delta}{\rho(1+\delta)}$$

Check for $\delta=0$ makes sense → yes

$X(s) = X_B(s) + \delta D(s)$ solutions with

$$X_B'' + [K_x(s) + \Delta K_x]X_B = 0$$

$$D'' + [K_x(s) + \Delta K_x]D = \frac{1}{\rho} + O(\delta) \quad \text{neglect}$$

With

$$K_x = \frac{1}{\rho^2} - K(s)$$

← Here Lee changes
definition

$$K(s) = B_1/B\rho$$

instead of -...

$$\Delta K_x = \left[\frac{-2}{\rho^2} + K(s) \right] \delta + O(\delta^2)$$

↳ Chromatic term, neglect now, return
to later

- Piece-wise constant description within a given element (like Lec 3)
 $\Rightarrow$ matrix method to solve

$$\begin{bmatrix} D(s) \\ D'(s) \end{bmatrix} = M(s|s_0) \begin{bmatrix} D(s_0) \\ D'(s_0) \end{bmatrix} + \begin{bmatrix} d \\ d' \end{bmatrix}$$

$\uparrow$ For homogeneous term $\uparrow$ particular solution

$$\bar{d} = \begin{bmatrix} d \\ d' \end{bmatrix} \text{ dispersion vector}$$

Use extended matrix form
 (Lec 2, III.2, also useful for linear errors, showed but did not discuss in Lec 5)

$$\begin{bmatrix} D(s) \\ D'(s) \\ 1 \end{bmatrix} = \begin{bmatrix} M(s|s_0) & \bar{d} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} D(s_0) \\ D'(s_0) \\ 1 \end{bmatrix}$$

⇒ Extended transfer matrix

$$\begin{bmatrix} m_{11} & m_{12} & d \\ m_{21} & m_{22} & d' \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{aligned} m_{11} &= C(s) \\ m_{12} &= S(s) \\ m_{21} &= C'(s) \\ m_{22} &= S'(s) \end{aligned}$$

• Form of $\bar{d}$ for K regimes:

$$K=0 \text{ drift space} \Rightarrow \bar{d} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \text{quadrupole}$$

★ why? not introducing further dispersion

Think about this for closed-orbit solution $x(s) = \delta D(s)$ (perturbed closed orbit)

$$\Rightarrow \text{Drift space } M = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix}, \quad \bar{d} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} D(s) \\ D'(s) \\ 1 \end{bmatrix}_{s=l} = \begin{bmatrix} 1 & l & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} D_0 \\ D'_0 \\ 1 \end{bmatrix}_{s=s_0}$$

$$D(l) = D_0 + D'_0 l$$

$$x = \delta D$$

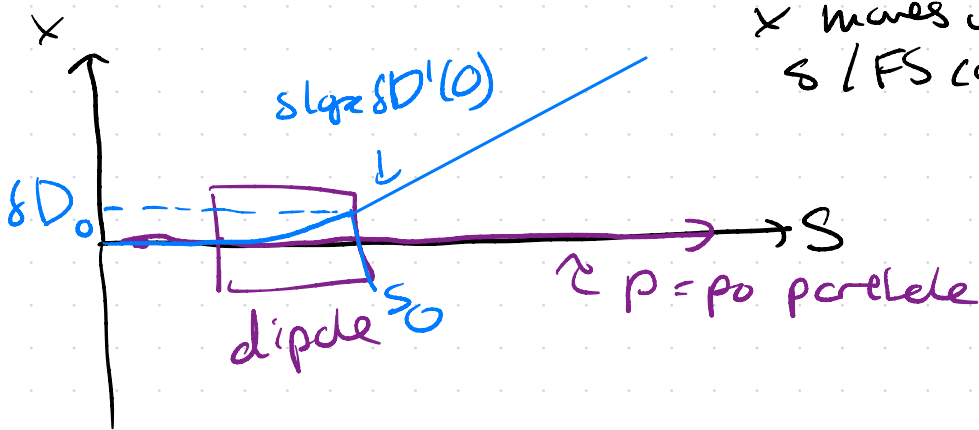
$$D'(l) = D'(0)$$

$$\Rightarrow x(l) = \delta D_0 + \delta D'_0 l$$

Perturbed orbit evolves but is not further perturbed

Helps to visualize kick + drift?

* remember,
x moves with
 δ / FS coord.



Quad case:

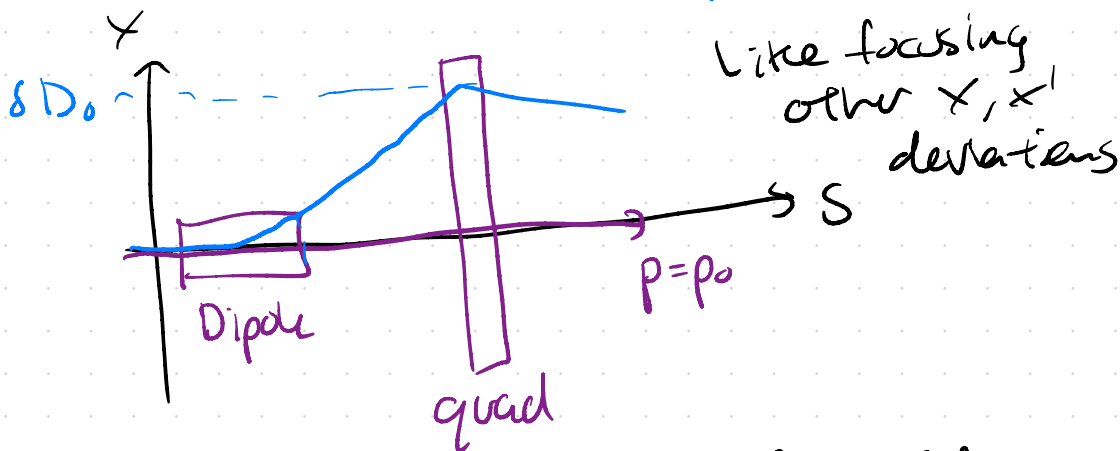
$$M = \begin{bmatrix} 1 & 0 & 0 \\ -1/f & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} D \\ D' \\ 1 \end{bmatrix} = M \begin{bmatrix} D_0 \\ D'_0 \\ 1 \end{bmatrix} \Rightarrow$$

$$D(s) = D_0$$

$$D'(s) = -\frac{1}{f} D_0 + D'_0$$

changes trajectory
slope, focusing

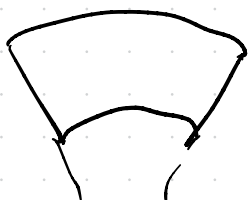


Focusing strength $\Rightarrow f$ differs
 $\hookrightarrow$ this is ΔK term well
 come back to

compare these with sector bend

dipole $K_x = \frac{1}{\rho^2}$

$$\bar{d} = \begin{bmatrix} \rho(1 - \cos\theta) \\ \sin\theta \\ 1 \end{bmatrix}$$



$$M = \begin{bmatrix} \cos \theta & p \sin \theta & p(1 - \cos \theta) \\ -\frac{\sin \theta}{p} & \cos \theta & \sin \theta \\ 0 & 0 & 1 \end{bmatrix}$$

Thin ($\theta \ll 1$) $\Rightarrow M = \begin{bmatrix} 1 & l & \frac{1}{2} l \theta \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{bmatrix}$ *

* From Taylor expansion

$$p = \frac{l}{\theta} \quad \cos \theta \rightarrow 1 - \frac{\theta^2}{2} + \dots \quad \text{don't (usually) include } O(\theta^2)$$

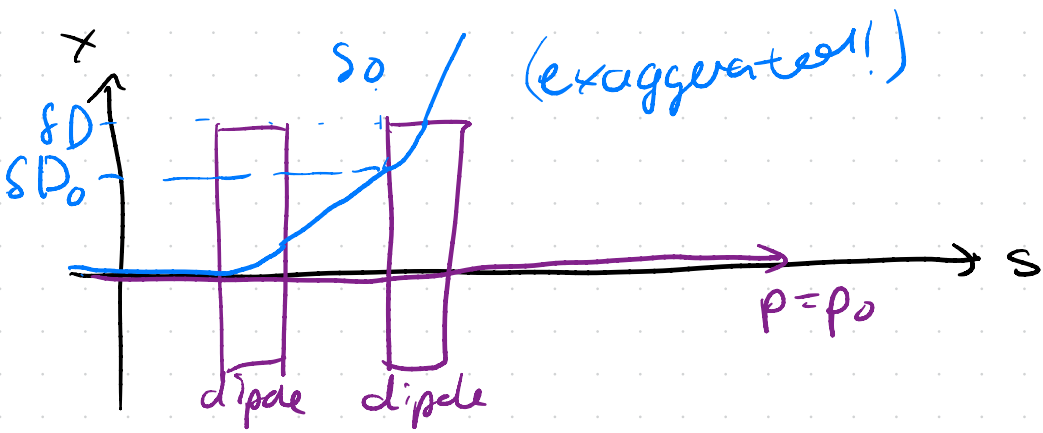
$$\frac{l}{\theta} \sin \theta \rightarrow \underline{l} - \frac{l \theta^2}{6} + \dots$$

$$\frac{l}{\theta} (1 - \cos \theta) \rightarrow \underline{\frac{l \theta}{2}} - \frac{l \theta^3}{24} \dots \quad \text{first terms only}$$

$$\begin{bmatrix} D \\ D' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & l & \frac{1}{2} l \theta \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} D_0 \\ D'_0 \\ 1 \end{bmatrix}$$

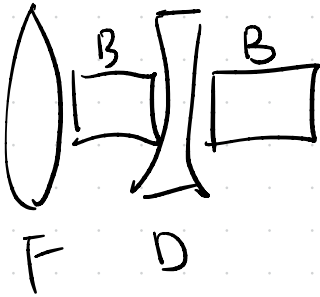
$$\Rightarrow D = D_0 + l D'_0 + \frac{1}{2} l \theta \quad D \text{ increases}$$

$$D' = D'_0 + \theta \quad D' \text{ increases}$$

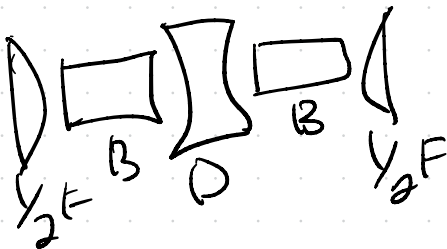


general piecewise form: ~~slide~~

Example: FODO cell + dipole bends



$$M = M_B M_D M_B M_F$$



$$M = M_{F/2} M_B M_D M_B M_{F/2}$$

⇒ Apply periodicity, find

$$D_F = \frac{L_1 \Theta (1 + \frac{1}{2} \sin(\Phi/2))}{\sin^2(\Phi/2)}, \quad D'_F = 0$$

$\leftarrow$ cell length
 $\uparrow$
 at focusing quad

Φ phase advance
 per period

$$D_D = \frac{L_1 \Theta (1 - \frac{1}{2} \sin(\Phi/2))}{\sin^2(\Phi/2)}, \quad D'_D = 0$$

$D_F > D_D$, $D' = 0$ Boch cases
 $\hookrightarrow$ min/max of D

• To consider linear magnet errors, same process as yesterday, perturbation in diff. eq. for $D(s)$ ~~slide~~

Momentum compaction factor

• Dispersion from $p \neq p_0$ influences transverse motion, which in turn changes the path length through accelerator

C ideal circumference

$$\Delta C = \left[\oint \frac{D(s) ds}{\rho} \right] \delta$$

$$\alpha_c \equiv \frac{1}{C} \frac{d\Delta C}{d\delta} = \frac{1}{C} \oint \frac{D(s) ds}{\rho} \Rightarrow \text{Discussed already}$$

↑
momentum compaction
factor

Impact of dispersion

Always have some Δp of overall beam
↳ bends will introduce dispersion

$$x(s) = x_B(s) + \delta D(s)$$

↳ impacts beam size

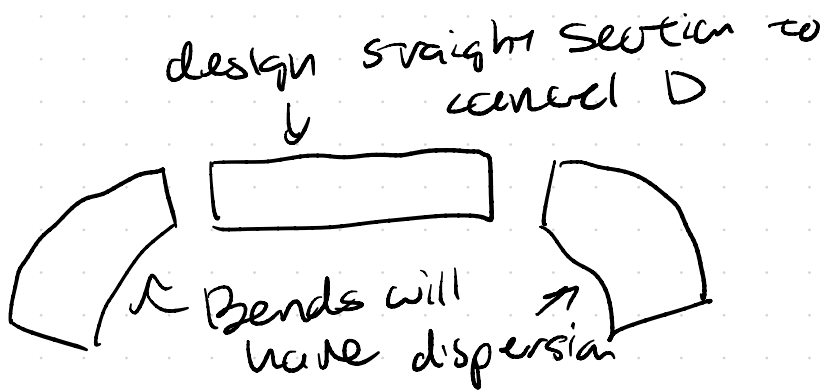
$$\sigma_x^2(s) = \beta_x(s) \epsilon_{x, rms} + D^2(s) \left\langle \left(\frac{\Delta p}{p_0} \right)^2 \right\rangle$$

Reduce $D \rightarrow$ reduce beam size

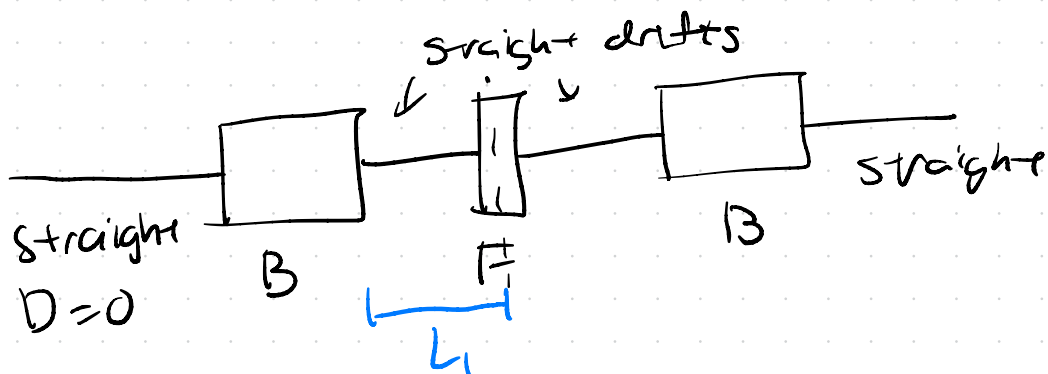
- $D=0$ case: beam size only determined by betatron oscillations

Dispersion Matching Lee 2.IV.4-5

- Designing lattice to match regions, eliminate dispersion at certain points
- still have Δp , but at some point in space reproduce $x(s) = x_B(s)$
↳ "achromats"



e.g. Double Bend Achromat (DBA)



At center of quadrupole

$$\begin{bmatrix} D_c \\ 0 \\ 1 \end{bmatrix} = M \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(A blue arrow points from the text $D' = 0 \Rightarrow D_c \text{ at max}$ to the first element of the left vector.)
 (A blue arrow points from the text 'zero initial dispersion' to the top element of the right vector.)
 * slide

$$f = \frac{1}{2} \left(L_1 + \frac{1}{2} L \right), \quad D_c = \left(L_1 + \frac{1}{2} L \right) \theta$$

(A blue arrow points from the text 'drift length' to L_1 .)
 (A blue arrow points from the text 'dipole length' to L .)

- Many other types + more complex designs! SY Lec 2. IV.5
HW 8.4.3, 14.3.1,

Chromatic aberration

Lee 2.IV
H.W. 15.4

- With $\delta \neq 0$ we have $x(s) = x_\beta(s) + \delta D(s)$
- Beta-tron oscillations around $\delta D(s)$ closed orbit

- Back to the term we ignored previously

$$x''_\beta + [K_x(s) + \Delta K_x] x_\beta = 0$$

$$D'' + [K_x(s) + \Delta K_x] D = \frac{1}{\rho} \quad B_1 = \frac{\partial B_z}{\partial x}$$

with $K_x = \frac{1}{\rho^2} - K(s) \quad K = \frac{B_1}{B\rho}$

$$\Delta K_x = \left[-\frac{2}{\rho^2} + K(s) \right] \delta + O(\delta^2) \approx -K_x \delta$$

$$\Delta K_z = -K(s) \delta + O(\delta^2) \approx -K_z \delta$$

$$\left(\begin{array}{c} \text{gradient} \\ \text{focusing} \\ \text{function} \end{array} \right) \cdot \left(\frac{\Delta p}{p} \right)$$

* Why negative?

$$\delta = \frac{\Delta P}{P} = \frac{P - P_0}{P_0} \quad \delta > 0 \quad \text{particle has } p > p_0$$

$$\Rightarrow K = \frac{B_1}{B\rho} \quad \uparrow P \Rightarrow \uparrow \rho \quad (\text{fixed gradient } B_1)$$

$\Rightarrow$ less effective focusing
(underfocused)

- opposite for $\delta < 0 \Rightarrow$ overfocused
- ΔK can include many other terms, e.g. dipole edge effects (soon!)
- Recall Lec 5: quadrupole gradient errors lead to tune shifts + modulation of β function
 - $\hookrightarrow$ same in case of chromatic aberration from δ

Tune Shift:

$$\Delta \nu_x = \frac{1}{4\pi} \oint \beta_x \Delta k_x ds \approx \left(-\frac{1}{4\pi} \oint \beta_x k_x ds \right) \delta$$

same form for z

Chromaticity $C_x \equiv \frac{d(\Delta \nu_x)}{d\delta}$

Can arise from many things, so we also define "natural chromaticity",

$$C_{x, \text{nat}} \approx -\frac{1}{4\pi} \oint \beta_x k_x ds$$

* all same for z

Solely from quadrupoles as designed

$$\delta > 0 \Rightarrow \Delta \nu_x < 0, C_x < 0$$

Momentum spread $\rightarrow$ betatron tune spread, can lead to hitting resonances

RHIC #s:

entire range of Lec 5 tune maps! 

$$C_{y, \text{nat}} \approx -50, \delta = \pm 5 \times 10^{-3} \Rightarrow \Delta \nu \approx 0.5$$

Requires correction for storage ring

↳ sextupoles $n=3$

$$\frac{\Delta B_z}{B\rho} = \frac{B_2}{2B\rho} (x^2 - z^2)$$

quadratic in coordinates

$$\frac{\Delta B_y}{B\rho} = \frac{B_2}{B\rho} xz$$

* Recall Lec 5
multiple expansion

$$B_2 = \left. \frac{\partial^2 B_z}{\partial x^2} \right|_{x=z=0}$$

• Define effective sextupole focusing strength:

$$S(s) = -\frac{B_2}{B\rho}$$

• Incorporate $x(s) = x_\beta(s) + \delta D(s)$

$$\frac{\Delta B_z}{B\rho} = -\left(S(s)D(s)\delta\right)x_\beta - \frac{S(s)}{2}(x_\beta^2 - z_\beta^2) - \frac{S(s)}{2}D^2(s)\delta^2$$

$$\frac{\Delta B_x}{B\rho} = -[S(s)D(s)\delta]z_\beta - S(s)x_\beta z_\beta$$

- Key takeaways:

- effective quad focusing

$$\Delta K_x = S(s)D(s)\delta, \quad \Delta K_z = -S(s)D(s)\delta$$

→ Use Sextupoles for chromaticity correction

- BUT $x_\beta^2 - z_\beta^2$ term can produce non-linear resonances

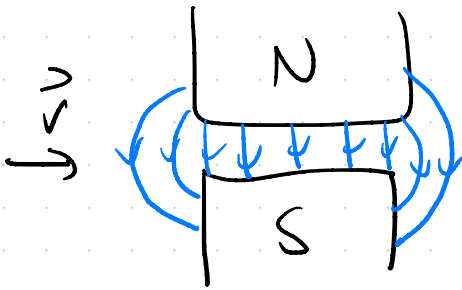
↳ SY Lee 2.VII.1

table 2.3 & slides

Dipole edge effects

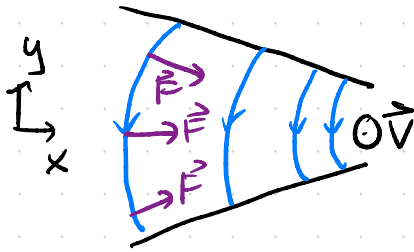
H.W. 7.3.2

H.W. 7.3.5



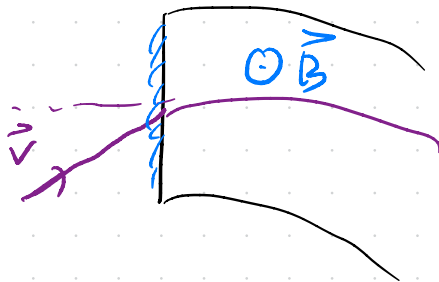
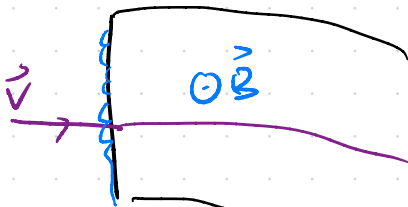
Gradient at edges
in both directions

Recall notation:



vertical focusing
horizontal defocusing

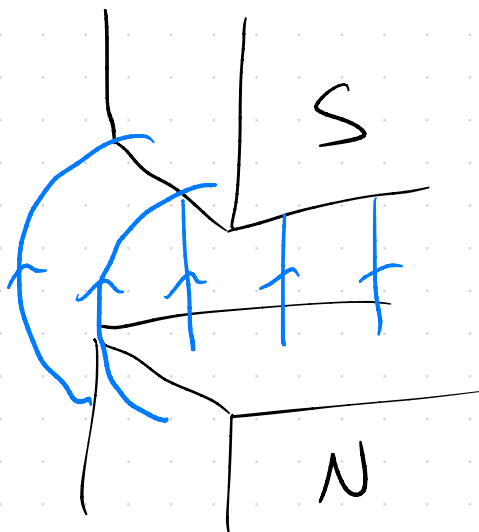
Particle entering at an angle w.r.t. pole face
in presence of gradient



top view

side view

base $\hat{x}, \hat{y}, \hat{z}$
to simplify
picture



$$\hat{x} \uparrow \hat{y} \uparrow \hat{z}$$

Ideal field:

$$\vec{B} = B_0 \hat{y} \quad \text{"hard edge"}$$

With gradient at edge

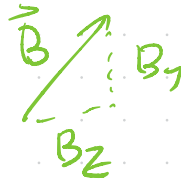
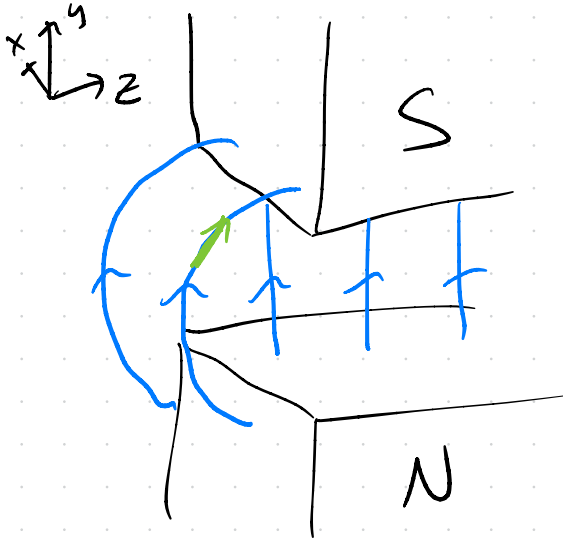
$$\vec{\nabla} \times \vec{B} = 0$$

$$= \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{x} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{y} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \hat{z}$$

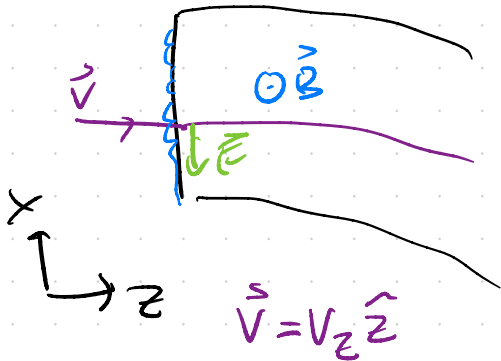
Focus on center region, no variation in x
(as drawn) (i.e. dipole is wide compared to
the beam) so no $\hat{y}$ and $\hat{z}$ terms cancel

$$\Rightarrow \text{edge: } \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial x} = 0$$

Now same B_z !
(as drawn)



Back to top now:



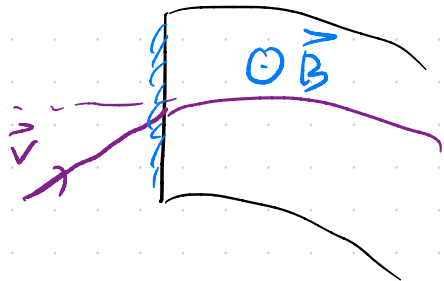
$$\vec{v} = v_z \hat{z}$$

$$\vec{F} = q \vec{v} \times \vec{B} \Rightarrow$$

$$\vec{v} \times \vec{B} = (v_z \hat{z}) \times (B_y \hat{y} + B_z \hat{z})$$

$$= v_z B_y (-\hat{x}) \quad \text{bending}$$

no impact from B_z



$$\vec{v} = v_x \hat{x} + v_z \hat{z}$$

$$\vec{v} \times \vec{B} = (v_x \hat{x} + v_z \hat{z}) \times (B_y \hat{y} + B_z \hat{z})$$

$$\Rightarrow \underline{v_x B_y \hat{z}} + v_z B_y (-\hat{x})$$

$$+ \underline{v_x B_z (-\hat{y})}$$

